

Self-hosted vs managed cloud: A comparison

Factor	Self-hosted	Managed cloud
Control	You have full control over the OS, hardware, and networking. You configure what you want and how you want.	Locked to what the provider offers. It works for most teams, but don't expect kernel-level tweaks.
Cost model	Higher upfront (hardware, contracts). Predictable after that. Cheaper long-term at scale.	Pay-as-you-go sounds great until the bill arrives. Flexible, but hard to forecast accurately.
Scalability	Manual. You provision, configure, and expand. It needs proper planning.	Auto-scaling is built in. Handles traffic spikes without you having to manage anything. Big win for unpredictable workloads.
Security	Entirely your problem. Firewalls, patching, and access controls are all on your team. No one to blame but yourself.	Shared responsibility. Provider secures the underlying infrastructure; you secure your app and data. Easier baseline, but don't get complacent.
Maintenance	Your team handles everything such as patches, upgrades, hardware failures. It needs solid automation, or it gets painful fast.	Most operational tasks are automated. Less toil and a smaller ops team are needed.

self-hosted and managed cloud infrastructure. They help you easily replicate a lot of cloud functionality on your own infrastructure.

Open source tools include MinIO for object storage, Supabase as a backend platform, Dokku or CapRover for PaaS-style deployments, and OpenFaaS for serverless workloads. Combined with Terraform or Pulumi for Infrastructure as Code, a competent team can build cloud-like capabilities without paying cloud-like prices.

This is genuinely changing the scenario for mid-sized companies.

Instead of choosing between self-hosting and managed cloud, many organisations prefer to choose hybrid infrastructure strategies. For example, they use self-hosted infrastructure for core databases and managed cloud for large-scale data processing.

Multi-cloud strategies take this further by distributing workloads

across providers to reduce single-vendor dependency. Kubernetes makes this more practical by providing a consistent deployment layer across different environments.

If you opt for a self-hosted cloud, ensure you follow these practices:

Automate your backups: Restic and BorgBackup are ideal tools for automating backup. They store encrypted copies in remote locations. So, don't skip this.

Set up proper monitoring: Prometheus + Grafana give you real-time visibility into system health. They give an alert before they impact the user.

Centralise your logs: ELK stack or Grafana Loki allow engineers to

analyse logs across distributed systems. Distributed systems without centralised logging are a debugging nightmare.

Tighten your security: Your infrastructure security should include strict firewall policies, encrypted communication, and strong authentication. Use WireGuard or Tailscale for secure remote access. Don't expose management interfaces to the public internet.

What's coming next?

A few trends are worth watching.

- Edge computing is pushing computing closer to end users. K3s makes it practical to run lightweight clusters at the edge that sync back to central systems.
- Platform engineering is growing. Many companies are building internal developer platforms to simplify infrastructure management for development teams. Backstage is the popular tool for this. It lets developers deploy without managing infrastructure directly.
- Decentralised storage is still early, but peer-to-peer storage architectures could eventually challenge traditional cloud storage economics.

There's no one-size-fits-all answer here. Managed cloud wins on speed, simplicity, and operational overhead. Self-hosting wins on control, privacy, and long-term cost at scale.

For most teams, the smart move is a hybrid approach. They should opt for the managed cloud where convenience matters, and go for self-hosted where control or cost efficiency matters more.

Choose what fits your workload, your team's capabilities, and your budget. Don't default to either extreme without running the numbers. **END** 🐧

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